

Helix OTA — Continuation / Resume Handoff

Field	Value
Revision	2
Created	2026-06-07
Last modified	2026-06-08
Status	active — resume with “continue”
Status summary	Single source of truth for resuming work. Captures exactly what is DONE (verified), the git state, and the prioritized NEXT steps. Everything below is committed to main and pushed to all 4 upstreams (GitHub, GitLab, GitFlic, GitVerse).

How to resume

Type **continue** in a new session. Read this file + the memory index first. All work is on main (latest commit pushed to all upstreams). Working branch history: foundation → research → MVP specs → submodule impls → server, each merged to main.

Locked decisions (do not re-litigate)

- Stack: **Go + Gin + Brotli + HTTP/3(QUIC)→HTTP/2 fallback, REST primary**; reuse vasic-digital/http3.
- Strategy: **native Android A/B (update_engine + AVB/dm-verity + auto-rollback) + custom Go control plane**; wrap OSS only where it adds value (hawkBit gated, ADR-0001).
- Trust: signing + SHA-256 + AVB for MVP; TUF server-side then device-side in 1.0.1 (ADR-0002).
- Topology: modular monolith for MVP (ADR-0003).
- New submodule repos auto-created PUBLIC on GitHub + GitLab (pre-authorized).
- docs/research/main_specs/additions/ files are authoritative input — always analyze + fold in.
- Commit + push to ALL upstreams regularly. Merge to main when a milestone is done.

Session 2026-06-08 update (DONE, on main, pushed all 4 upstreams)

- **Constitution submodule relocated** HelixConstitution/ → **constitution/** (repo/URL unchanged) via `git mv`; fixed the only filesystem path refs (`tests/test_strategy.md`, `additions/initial_research_01.md`) incl. their pre-existing off-by-one link depth. Prose “HelixConstitution” clause citations (the repo NAME) intentionally left.
- **Inheritance wired**: parent `CLAUDE.md`, `AGENTS.md`, `docs/guides/HELIX_OTA_CONSTITUTION.md` inherit from the submodule; `tests/inheritance_gate.sh` (5 invariants, matches the exact §11.4 forensic-anchor heading) + `tests/test_constitution_inheritance.sh` (gate green clean + §1.1 paired-mutation proven via `constitution/meta_test_inheritance.sh` + recursive submodule pointer check) + `tests/pre_build_verification.sh` + installed pre-commit hook. All 6 owned `ota-*` submodules carry `CLAUDE.md`+`AGENTS.md` inheritance pointers (pushed github+gitlab; gitlinks bumped to `v0.1.0-1-g...`). Commits `ee5dc7d`, `e11a221`.
- **All 3 additions processed** (operator mandate — nothing skipped): per-input exhaustive inventories in `research/additions_analysis/0{1,2,3}_analysis.md`; `research/additions_synthesis.md` → **Rev 2** with consolidated 14-gap register (§8), new-work routing (§9), addition-#3 conflict reconciliation (§10, locks `Gin/ed25519/JWT/releases+deployments` ≥90%), future-phase catalogue (§11), UNVERIFIED register (§12). MVP-critical **G1 (anti-downgrade invariant)** implemented as regression tests; `handleClientUpdate` 79.3%→86.2%. Commit `629b4eb`.
- **Constitution path is now constitution/** — update any new references accordingly.

Server hardening — DONE this session (handoff NEXT #1, real evidence)

- **Coverage**: upload handler 71.2%→**96.6%**, client-update 79.3%→**93.1%** (≥90% safety floor). Commit `64c3bfd`.
- **pgx PostgreSQL Repository** (`server/internal/store/postgres.go`): full `store.Repository` impl; shared `contract_test.go` proves parity with memory; **integration test boots real Postgres via the containers submodule on podman** (`go test -tags integration ./internal/store/`), evidence in `docs/qa/20260608-pgx-postgres-integration/`. Surfaced+fixed a real idempotency overwrite bug. Commit `96fdecb`.
- **Brotli→gzip→identity** compression middleware (`server/internal/api/compression.go`). Commit `b26f30e`.
- **HTTP/3 (QUIC)+HTTP/2 fallback** via the new submodules/`http3` (`digital.vasic.http3`) — `server/internal/transport/`; wired into `cmd/ota-server` (TLS via `HELIX_TLS_CERT/KEY` → HTTPS port 8443; plain HTTP otherwise). Real `h3+h2` client test; evidence in `docs/qa/20260608-http3-h2-brotli-transport/`. Commit `1469edf`.

Parallel-wave deliverables — DONE this session (commit `8efb6b8`)

6 subagents closed additions gaps: **G2** validator hash-before-signature VERIFIED (FACT, file:line); **G7** staged-rollout engine spec + `migration_002` design; **G6** dashboard design; **G3/G4/G5** operational endpoints spec (`audit/telemetry-reads/`

group CRUD + proposed repo methods); **G10** CI (.github/workflows/ci.yml + CODEOWNERS + dependabot); **§11** future-phase folding (1.0.2-rollback, 1.0.3-delta-updates created; 1.X-linux/windows/other-os extended).

Operational endpoints — DONE this session (G3/G4/G5, real-Postgres-verified, memory+pgx parity)

- **G3 audit** (commit 9a703bf): auditMiddleware records successful mutating actions (reads/failures skipped, ids never leak into the action) + GET /audit (admin) + audit_logs table.
- **G4 telemetry reads** (commit eadaa7f): GET /devices/{id}/telemetry (device reads only its own) + GET /telemetry/overview (fleet counts).
- **G5 device-group CRUD** (commit e3e1307): full /groups CRUD + membership; writes operator/admin, delete admin-only. All three added store.Repository methods on memory + pgx, extended the shared contract, and pass the pgx integration test on real Postgres.

Staged rollout + more — DONE this session

- Migration 002 real-DB validated (e14942a). **Go engine** wired via the ota-rollout-engine brick: server/internal/rollout + REST POST/GET /deployments/{id}/rollout + /evaluate (create→start→advance→complete, halt-on-error-breach); 7 tests green (046ea07).
- End-user rollback reconciled INTO **1.0.1** per operator decision; 1.0.2-rollback superseded→folded (a4c2b3d).
- **G11** repo audit: all 6 ota-* PUBLIC on GitHub+GitLab (dab2f0e). **G12** NFR/load harness real measured percentiles (6547c7f).

Round 2 deliverables — DONE this session

- **Dashboard** scaffold (G6) — Vite+React+TS, real tsc/vite build exit 0 (813a00e).
- **6 submodule READMEs** enriched + pushed to each github+gitlab (813a00e).
- **Dependabot ×4 merged + verified**: gin 1.12.0, quic-go 0.60.0 (http3 transport re-verified), actions v6 (33039b8).
- **Rollout pgx StoragePort** — real-DB tested via containers submodule (8190e92).
- **§11.4.65 exports** for 13+2 new docs (242ee2f, a288a4b). **device_tuf.md** + **rollback_ux.md** specs (1.0.1).

Round 3 deliverables — DONE this session

- **Rollback-history store layer** (AppendRollback/ListRollbacks) memory+pgx, real-DB parity (e4165a6).
- **Recall endpoint** POST /deployments/{id}/recall + GET /rollbacks (c98dfac) — records rollback_history, validates deployment/target-release, operator/admin; 3 tests.
- **Delta-updates 1.0.3** design (ADR-0005 Option B) + **as-built endpoint reference** (server.go route table) + submodule README §11.4.65 siblings (573779a).

- **Autonomous e2e challenge** tests/e2e/challenge_operational.sh — real live-server run, 28 passed/0 failed/1 skip, independently re-verified (bb332a4).

Round 4 deliverables — DONE this session

- **pgx wired into main** (HELIX_DATABASE_URL): production uses the pgx repo + rollout StoragePort; **e2e challenge PASS (28/0/1) against a real Postgres-backed server** (1cdad81, docs/qa/20260608-pgx-server-e2e/).
- **migration 004** delta_artifacts real-DB validated; **CI e2e+loadtest jobs**; **README** doc-map §10; **HelixQA bank**; **threat-model** extended (6a3e213, c-threat).

Round 5 deliverables — DONE this session

- **Recall = forward-fix** (operator decision honor-AVB, 4e35c3e): handleRecall supersedes the current deployment + creates a NEW active deployment of the target release; the update-check anti-downgrade invariant means AVB is honored by construction. New store UpdateDeployment (memory+pgx, real-DB parity). Decision recorded in rollback_ux.md Rev 2 + threat_model.md §11.11 RESOLVED (f5ec504).

Round 6 deliverables — DONE this session

- **Delta-artifact store + API** (66464d7): delta_artifacts (base≠target CHECK + UNIQUE pair) on memory+pgx; POST/GET /deltas register+lookup; real-DB parity.
- **device-TUF client-decision memo** (recommend gomobile-go-tuf/v2, ADR-0002 §4.3) + sibling.
- **Additive WIDENs — full set landed** (operator-approved): audit ?since/? until filters + telemetry failure_rate/by_state (028e656, memory+pgx real-DB parity) + group member_count (4cb86d7). spec_impl_alignment.md Rev 3: rows 2+5+6-partial landed; as-built doc re-synced (recall now WIRED, /deltas + recall + rollbacks documented).

Round 7 — breaking WIDENs landed (operator ruling: WIDEN-impl)

- **Group** (a91271b + 02ad2d0): id→group_id; batch member-add → 200 {added, already_member, not_found}. Server+tests+e2e(36/0/1)+dashboard+bank+docs.
- **Audit** (fbaefbe): actor → object {user_id, subject}. Server+test+dashboard+docs.
- **Telemetry per-device** (2a48ab5): events→items, newest-first, ?limit/? cursor+next_cursor. Server+tests+dashboard+docs.
- **GET/members** (3b4b1d8 + e2e): device_ids[] → items[] of {device_id, added_at}; store gained device_group_members.added_at + ListGroupMembersDetailed (memory+pgx, real-DB parity); e2e 39/0/1.
- Each: full default suite green, all wire consumers updated in lockstep, spec_impl_alignment.md rows 6/8/1/4-structural + GET-members RESOLVED.

WIDEN ruling status: COMPLETE except two legitimately-parked items: row-4 richer telemetry fields (duration_ms/bytes_transferred) **blocked on UNVERIFIED ingest** (event source must carry them first) + the per-device telemetry filters; group/members list pagination (row 7) **deferred** (groups bounded — memo's own recommendation). Both need either ingest work or an operator nudge; not autonomously actionable now.

NEXT wave (still open)

1. **Device-side Android** (the substantial remaining feature work, Kotlin ota-android-agent/ota-update-engine-bridge): (a) update-check delta-selection — map device current-version → base artifact → FindDelta (server-side hook) + device payload apply; (b) device-side TUF (gomobile-go-tuf/v2 per the decision memo, gated on an arm64 .so-size/JNI measurement first).
2. **Parked WIDEN bits** (operator nudge or ingest work): row-4 richer telemetry fields (blocked on UNVERIFIED ingest) + per-device filters; group/members list pagination (deferred).
3. CODEOWNERS GitHub handle; make vasic-digital/containers + HelixConstitution GitLab mirrors public (or document GitHub-canonical) per the G11 audit.

Carried-forward gaps register

See additions_synthesis.md §8/§9 (14 gaps; most now specced — implementation pending). Numbering decision: 1.0.1 = staged-rollout; rollback→1.0.2, delta→1.0.3.

DONE (verified, on main, pushed)

1. **Spec corpus** (docs/research/main_specs/): master design + ADRs index; research = scored landscape report + 12 stack notes + 5 ADRs (ADR-0001..0005) + additions synthesis; foundation = glossary, STRIDE threat model, submodule reuse map, doc standards, requirements traceability, export pipeline; **1.0.0-mvp** full specs (api+OpenAPI, database+migrations, security, server, client_android+snippets, deployment, tests, VALIDATION_EVIDENCE); 6 Mermaid diagrams; future-phase outlines (1.0.1, 1.X-linux/windows/other-os); corpus README.
2. **Validation evidence (real)**: OpenAPI redocly-valid; **migrations applied to live PostgreSQL** (12 tables up, clean down); k8s kubeconform 5/5; compose YAML valid; all 6 diagrams render (mmdc); corpus exports = 50 HTML + 50 DOCX + SVG/PNG (scripts/export_docs.sh). PDF needs a LaTeX engine (not installed).
3. **6 reusable submodules** (created on GitHub HelixDevelopment/ + GitLab helixdevelopment1/, scaffolded, wired under submodules/, **implemented v0.1.0, tagged, pushed both hosts**):
 - ota-protocol (Go, 99.3%), ota-telemetry-schema (Go, 98.9%), ota-artifact-validator (Go, 97.8%, ed25519+sha256), ota-rollout-engine (Go, 94.9%, halt-wins+deterministic cohorts)
 - ota-update-engine-bridge (Kotlin; :core 27 tests; :android builds an AAR)

- ota-android-agent (Kotlin; :core 36 tests; :android build pending KMP/AGP-on-Gradle-9.5 alignment)
- 4. **Submodules wired:** HelixConstitution + containers at root; 6 ota-* under submodules/.
- 5. **Control-plane server** (server/, Go module github.com/HelixDevelopment/helix_ota/server): Gin modular monolith wiring the 4 Go modules; full /api/v1 MVP endpoints (auth, devices, artifacts upload+validate S1–S6, releases, deployments all-targets, client/update 204|200, telemetry, health); **66 httpstest integration tests pass**; gofmt/vet/build clean.

NEXT (prioritized)

1. **Server hardening to spec:** raise internal/api coverage to $\geq 90\%$ on handleUploadArtifact + handleClientUpdate (error-path branches); add a **pgx** Repository implementation behind the existing store.Repository interface + testcontainers-go integration tests against real Postgres; wire vasic-digital/http3 for HTTP/3→HTTP/2 + Brotli.
2. **ota-android-agent :android build:** switch its :android module from Kotlin MPP to plain com.android.library + org.jetbrains.kotlin.android (like the bridge, which builds), OR pin Gradle 8.x + AGP 8.7.x via a wrapper; then produce the AAR.
3. **React dashboard** (dashboard/): reuse UI-Components-React, Dashboard-Analytics-React, Auth-Context-React; secure login, upload, deploy, fleet health.
4. **1.0.1 staged-rollout:** promote the outline to full specs + implement rollout API (phases, pause/resume/abort), device-side TUF, end-user rollback; migration 002_* (deployment_phases, rollouts, rollback_history, TUF metadata).
5. **AOSP integration:** build the agent + bridge as a system/priv-app in the Orange Pi 5 Max AOSP tree; on-device e2e (download→verify→A/B apply→reboot→post-boot; corrupt-slot→auto-rollback).
6. **Containerize** the dev stack via the containers submodule; CI running all validators (redocly, Postgres migrations, kubeconform, go test, gradle :core:test, export render).
7. **Per-submodule:** full README/docs/manuals + GitFlic/GitVerse mirrors + tag mirroring (§4).

Toolchain facts (this host)

Go 1.26.2; Android SDK (platforms 31–36.1, build-tools, NDK); Gradle 9.5 + Kotlin 2.3.20; **AGP 8.5.2 + plain kotlin.android works on Gradle 9.5; Kotlin MPP does NOT** (needs Gradle 8.x). pandoc/mmdc present; no LaTeX/drawio/plantuml/graphviz/docker. Live PostgreSQL server available. gh+glab authed. API rate limits are transient — keep waves ≤ 6 agents, don't overlap workflows.